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METHOD OF MANUFACTURING SEMICONDUCTOR DEVICE AND SEMICONDUCTOR DEVICE MANUFACTURING SYSTEM

Abstract

A method of manufacturing a semiconductor device includes supplying a developer on a substrate. A distribution information of the developer on the substrate is captured by a capturing device of an analysis subsystem. An assessment result is generated based on the distribution information by a processor of the analysis subsystem. Whether to perform an adjustment operation is determined based on the assessment result.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to a method of manufacturing a semiconductor device and a semiconductor device manufacturing system.

Description of Related Art

[0002] The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling-down has also increased the complexity of processing and manufacturing ICs.

[0003] A method of manufacturing a semiconductor device includes forming a resist layer on a substrate, a baking process, a developing process and other operations for creating two-dimensional (2D) and three-dimensional (3D) patterns in microelectronic devices. During the developing process, uneven dispense rate of the developer may cause poor quality of developed resist layer and also adversely affect a quality of the semiconductor device to be manufactured in a subsequent process.

[0004] In this regard, how to provide a method of manufacturing a semiconductor device and a semiconductor device manufacturing system that can overcome the aforementioned problem is one of the targets in the research and development in the related fields.

SUMMARY

[0005] One aspect of the present disclosure is a method of manufacturing a semiconductor device.

[0006] According to some embodiments of the present disclosure, a method of manufacturing a semiconductor device includes supplying a developer on a substrate. A distribution information of the developer on the substrate is captured by a capturing device of an analysis subsystem. An assessment result is generated based on the distribution information by a processor of the analysis subsystem. Whether to perform an adjustment operation is determined based on the assessment result.

[0007] In some embodiments, performing the adjustment operation includes changing a rotational speed of a motor, and the motor is connected to a substrate support stage below the substrate.

[0008] In some embodiments, performing the adjustment operation includes changing a thrust of a pump.

[0009] In some embodiments, the thrust of the pump is changed such that a dispense rate of the developer is changed.

[0010] In some embodiments, if it is determined not to perform the adjustment operation based on the assessment result, a rotational speed of a motor and a thrust of a pump are maintained, and in which the motor is connected to a substrate support stage below the substrate.

[0011] In some embodiments, the assessment result is generated by comparing the distribution information with a reference distribution information saved in the analysis subsystem.

[0012] In some embodiments, the method further includes transmitting a control signal based on the assessment result from the processor of the analysis subsystem.

[0013] In some embodiments, the method further includes rotating the substrate prior to capturing the distribution information of the developer on the substrate.

[0014] In some embodiments, the method further includes stopping supplying the developer after determining whether to perform the adjustment operation based on the assessment result.

[0015] In some embodiments, the distribution information of the developer on the substrate includes a surface image of the developer on the substrate.

[0016] In some embodiments, the method further includes prior to supplying the developer on the substrate, forming a resist layer over the substrate.

[0017] Another aspect of the present disclosure is a semiconductor device manufacturing system.

[0018] According to some embodiments of the present disclosure, a semiconductor device manufacturing system includes a chamber, a substrate support stage, a nozzle, and an analysis subsystem. The substrate support stage is positioned in the chamber and is configured to support a substrate. The nozzle is positioned over the substrate support stage and is configured to supply a developer on the substrate during a developing process. The analysis subsystem includes a capturing device and a processor. The capturing device of the analysis subsystem is configured to capture a distribution information of the developer on the substrate. The processor of the analysis subsystem is configured to generate an assessment result based on the distribution information, and the processor of the analysis subsystem is further configured to determine whether to perform an adjustment operation based on the assessment result.

[0019] In some embodiments, the semiconductor device manufacturing system further includes an adjustment controller connected to the analysis subsystem. The processor of the analysis subsystem is further configured to generate a first control signal and transmit the first control signal to the adjustment controller if it is determined to perform the adjustment operation.

[0020] In some embodiments, the adjustment controller is configured to change a rotational speed of a motor in response to the first control signal, and the motor is connected to the substrate support stage and configured to rotate the substrate.

[0021] In some embodiments, the adjustment controller is configured to change a thrust of a pump in response to the first control signal, and the pump is connected to the nozzle and configured to adjust a dispense rate of the developer from the nozzle.

[0022] In some embodiments, the semiconductor device manufacturing system further includes an adjustment controller connected to the analysis subsystem. The processor of the analysis subsystem is further configured to generate a second control signal and transmit the second control signal to the adjustment controller if it is determined not to perform the adjustment operation.

[0023] In some embodiments, the adjustment controller is configured to maintain a rotational speed of a motor and a thrust of a pump in response to the second control signal, the motor is connected to the substrate support stage and configured to rotate the substrate, and the pump is connected to the nozzle and configured to adjust a dispense rate of the developer from the nozzle.

[0024] In some embodiments, the processor of the analysis subsystem is configured to generate the assessment result by comparing the distribution information with a reference distribution information.

[0025] In some embodiments, the semiconductor device manufacturing system includes an adjustment controller and a main server. The adjustment controller is connected to the analysis subsystem. The connected main server is connected to the adjustment controller, and the main server is configured to start the developing process.

[0026] In some embodiments, the capturing device of the analysis subsystem includes at least one camera, and the distribution information of the developer on the substrate includes a surface image of the developer on the substrate.

[0027] In the aforementioned embodiments, since the analysis subsystem of the semiconductor device manufacturing system is configured to capture the distribution information of the developer and to generate the assessment result based the distribution information, the developer can be distributed more uniformly (or evenly) over the substrate. As such, a better developing process can be achieved and a quality of the semiconductor device can be improved.

[0028] It is to be understood that both the foregoing general description and the following detailed description are by examples, and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] The disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

[0030] FIG. **1** is a flow chart of a method of manufacturing a semiconductor device in accordance with some embodiments of the present disclosure.

[0031] FIG. **2** is a schematic view of a semiconductor device manufacturing system in accordance with some embodiments of the present disclosure.

[0032] FIG. **3** is a flow chart of a method of manufacturing a semiconductor device in accordance with some embodiments of the present disclosure.

[0033] FIGS. **4-6** are schematic views of the semiconductor device, in portion or entirety, at various fabrication stages in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0034] Reference will now be made in detail to the present embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the description to refer to the same or like parts.

[0035] As used herein, “around,” “about,” “approximately,” or “substantially” shall generally mean within 20 percent, or within 10 percent, or within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around,” “about,” “approximately,” or “substantially” can be inferred if not expressly stated.

[0036] FIG. **1** is a flow chart of a method **100** of manufacturing a semiconductor device (e.g., including a substrate and a resist layer thereon) in accordance with some embodiments of the present disclosure. In some embodiments, the method **100** is implemented, in whole or in part, by a system employing lithography processes, such as deep ultraviolet (DUV) lithography, extreme ultraviolet (EUV) lithography, e-beam lithography, x-ray lithography, and/or other lithography to enhance lithography resolution. For illustration, the method **100** will be described along with the drawings shown in FIGS. **2-7**. The method **100** is merely an example, and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional steps (or operations) can be provided before, during, and after the method **100**, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the process. For clarity and ease of explanation, some elements of the figures have been simplified.

[0037] At step **110**, a resist layer is formed over a substrate. In some embodiments, the resist layer is a positive tone resist. In some other embodiments, the resist layer is a negative tone resist.

[0038] The substrate (e.g., substrate **210** in FIG. **2**) may be a semiconductor substrate (e.g., a wafer), a mask (also called a photomask or reticle), or any base material on which processing may be conducted to provide layers of material to form various features of an IC device. Depending on an IC fabrication stage, the substrate includes various material layers (e.g., dielectric layers, semiconductor layers, and/or conductive layers) configured to form IC features (e.g., doped regions/features, isolation features, gate features, source/drain features (including epitaxial source/drain features), interconnect features, other features, or combinations thereof). In some embodiments, the substrate includes a semiconductor substrate, such as a silicon substrate. The substrate may include another elementary semiconductor, such as germanium; a compound semiconductor, such as silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor, such as SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

[0039] A resist layer (e.g., resist layer **220** in FIG. **2**) is formed over the substrate by a suitable process. For example, the resist layer is formed over the substrate by dropping a photoresist material through a nozzle. The resist layer is also called a photoresist layer, photosensitive layer, imaging layer, patterning layer, or radiation sensitive layer. The resist layer is sensitive to radiation used during a lithography exposure process, such as DUV radiation, EUV radiation, e-beam

radiation, ion beam radiation, and/or other suitable radiation.

[0040] In some embodiments, the resist layer includes a positive tone resist. When a developer is a positive tone developer (PTD), portions of the positive tone resist exposed to radiation become soluble to the developer, and unexposed portions of the positive tone resist remain insoluble to the developer. When a developer is a negative tone developer (NTD), portions of the positive tone resist exposed to radiation become insoluble to the developer, and unexposed portions of the positive tone resist remain soluble (or exhibit increased solubility) to the developer. For example, when the negative tone developer such as a hydrophobic organic solvent applied to the exposed portions of the positive tone resist, the unexposed (hydrophobic) portions of the positive tone resist are dissolved by the NTD and the exposed (hydrophilic) portions of the positive tone resist remain after the developing process to form a developed resist layer.

[0041] In some other embodiments, the resist layer includes a negative tone resist. When a developer is the positive tone developer, portions of the resist layer exposed to radiation become insoluble (or exhibit decreased solubility) to the developer, and unexposed portions of the negative tone resist remain soluble to the developer. When a developer is a negative tone developer (NTD), portions of the negative tone resist exposed to radiation become soluble to the developer, and unexposed portions of the negative tone resist remain insoluble to the developer.

[0042] At step **120** of FIG. **1**, a bake process is performed. In some embodiments, the resist layer is baked at an elevated temperature using a pre-exposure baking process to evaporate solvents in the resist layer. In some embodiments, the baking process is performed at a temperature suitable to evaporate the solvent, and the temperature depends upon the materials chosen for the resist layer. The baking process is performed for a time sufficient to cure and dry the resist layer.

[0043] At step **130** of FIG. **1**, an exposure process is performed. In some embodiments, the resist layer is exposed to radiation in the form of electromagnetic waves. In some embodiments, the radiation includes DUV radiation, EUV radiation, x-ray radiation, e-beam radiation, ion-beam radiation, and/or other suitable radiation. The exposure process can be in air or liquid, or vacuum. In some embodiments, the radiation is patterned using a mask having an IC pattern defined therein, such that the patterned radiation forms an image of the IC pattern on the resist layer. The mask transmits, absorbs, and/or reflects the radiation depending on the IC pattern, along with mask technologies used to fabricate the mask. In some embodiments, the radiation beam is patterned by directly modulating the radiation beam according to an IC pattern without using a mask (sometimes called maskless lithography).

[0044] At step **140** of FIG. **1**, a bake process is performed. In some embodiments, the exposed resist layer is baked at an elevated temperature using a post-exposure baking process. Specifically, after the exposure process, a post-exposure baking process may be performed on the resist layer.

[0045] At step **150** of FIG. **1**, a developing process is performed. FIG. **2** is a schematic view of a semiconductor device manufacturing system **200** in accordance with some embodiments of the present disclosure, and FIG. **3** is a flow chart of the step **150** of the method **100** in FIG. **1** in accordance with some embodiments of the present disclosure. The step **150** may include multiple steps/operations **151-157** (see FIG. **3**) and can be performed by using the semiconductor device manufacturing system **200** shown in FIG. **2**. For illustration, the flow chart of the step **150** will be described along with the drawings shown in FIGS. **2** and **3**. The flow chart shown in FIG. **3** is merely an example, and is not intended to limit the present disclosure beyond what is explicitly recited in the claims. Additional operations can be provided before, during, and after the step **150**, and some steps described can be replaced, eliminated, or moved around for additional embodiments of the process. For clarity and ease of explanation, some elements of the figures have been simplified.

[0046] Reference is made to FIG. **2**. The semiconductor device manufacturing system **200** includes a chamber **205**, a substrate support stage **230**, a nozzle **240**, and an analysis subsystem **250**. The substrate support stage **230** is positioned in the chamber **205** and configured to support a substrate

210 (e.g., silicon wafer) and a resist layer **220** on the substrate **210**. The nozzle **240** is positioned in the chamber **205** and over the substrate support stage **230**, and the nozzle **240** is configured to supply a developer DP on the substrate **210** (or the resist layer **220** on the substrate **210**) during a developing process. The analysis subsystem **250** includes a capturing device **251** and a processor **253**. The capturing device **251** of the analysis subsystem **250** is configured to capture a distribution information of the developer DP on the substrate **210** (or the resist layer **220**). The processor **253** of the analysis subsystem **250** is configured to generate an assessment result based on the distribution information, and the processor **253** of the analysis subsystem **250** is further configured to determine whether to perform an adjustment operation based on the assessment result. With such configuration, the developer DP can be distributed more uniformly (or evenly) over the substrate **210** (or on the resist layer **220**). As such, the better developing process can be achieved and a quality of the semiconductor device can be improved.

[0047] In some embodiments, the semiconductor device manufacturing system **200** further includes an adjustment controller **260**, a motor **270**, a pump **280**, and a developer source **290**. The adjustment controller **260** is connected to the analysis subsystem **250**, the motor **270**, and the pump **280**. The motor **270** is connected to the substrate support stage **230** and the adjustment controller **260**, and the motor **270** is configured to rotate the substrate support stage **230** and also the layers (i.e., the substrate **210** and the resist layer **220**) on the substrate support stage **230**. The pump **280** is connected to the nozzle **240**, the adjustment controller **260**, and the developer source **290**, and the pump **280** is configured to adjust a dispense rate of the developer DP from the nozzle **240**. The developer source **290** is connected to the nozzle **240** in the chamber **205** through a gas line, and the developer source **290** is configured to provide the developer DP.

[0048] In some embodiments, the adjustment controller **260** is configured to adjust/change a rotational speed of the motor **270** such that a distribution of the developer DP on the substrate **210** is changed. For example, by increasing the rotational speed of the motor **270**, the developer DP flowing from the nozzle **240** spreads over the entire substrate **210** faster; and by decreasing the rotational speed of the motor **270**, the developer DP flowing from the nozzle **240** spreads over the entire substrate **210** slower. The adjustment controller **260** is further configured to adjust/change a thrust of the pump **280** such that a distribution of the developer DP on the substrate **210** is changed. For example, by increasing the thrust of the pump **280**, a dispense rate (or a flow) of the developer DP flowing from the nozzle **240** is increased such that the developer DP spreads over the entire substrate **210** faster; and by decreasing the thrust of the pump **280**, the dispense rate (or the flow) of the developer DP flowing from the nozzle **240** is decreased such that the developer DP spreads over the entire substrate **210** slower. As a result, by adjusting the rotational speed of the motor **270** and/or the thrust of the pump **280** through the adjustment controller **260**, the developer DP can be distributed more uniformly over the substrate **210**.

[0049] In some embodiments, the processor **253** of the analysis subsystem **250** is further configured to generate a first control signal and transmit the first control signal to the adjustment controller **260** if it is determined to perform the adjustment operation. The adjustment controller **260** then changes the rotational speed of the motor **270** and/or the thrust of the pump **280** in response to the first control signal. In some embodiments, the processor **253** of the analysis subsystem **250** is a central processing unit (CPU), a controller, or other analytical devices with analysis capabilities.

[0050] In some embodiments, the processor **253** of the analysis subsystem **250** is further configured to generate a second control signal and transmit the second control signal to the adjustment controller **260** if it is determined to not perform the adjustment operation. The adjustment controller **260** then maintain (i.e., does not change) the rotational speed of the motor **270** and/or the thrust of the pump **280** in response to the second control signal.

[0051] In some embodiments, the processor **253** of the analysis subsystem **250** is configured to generate the assessment result by comparing the distribution information with a reference

distribution information. The reference distribution information may be a uniform distribution information of the developer on the substrate analyzed previously. By comparing the distribution information with the reference distribution information, non-uniform (or uneven) distribution of the developer DP can be identified and thus the assessment result is generated.

[0052] If the processor **253** of the analysis subsystem **250** determines (or assesses) that the distribution information is not similar to the reference distribution information, then the assessment result indicates that the distribution information of the developer DP is not uniform. As a result, the processor **253** of the analysis subsystem **250** determines to perform the adjustment operation. On the other hand, if the processor **253** of the analysis subsystem **250** determines (or assesses) that the distribution information is similar to the reference distribution information, then the assessment result indicates that the distribution information of the developer DP is uniform. As a result, the processor **253** of the analysis subsystem **250** determines not to perform the adjustment operation. In some embodiments, the reference distribution information is saved in a storage unit (e.g., DRAM, a mechanical disk, a solid-state drive, or other types of storage medium as deemed appropriate) of the analysis subsystem **250**.

[0053] In some embodiments, the capturing device **251** of the analysis subsystem **250** includes at least one camera (e.g., one to ten cameras). The capturing device **251** may have high-speed camera lenses (e.g., greater than 10000 frame per second). The capturing device **251** of the analysis subsystem **250** is configured to capture multiples images (e.g., 3 to 10 images) in one second. In some embodiments, the distribution information of the developer DP on the substrate **210** includes a surface image of the developer DP on the substrate **210**.

[0054] In some embodiments, the processor **253** of the analysis subsystem **250** is configured to generate the assessment result using AI-based image recognition scheme. In some embodiments, the processor **253** of the analysis subsystem **250** is configured to use the distribution information (e.g., surface image) captured by the capturing device **251** as an input to a function generated by running a machine learning algorithm on a training data set. The training data set may include multiple training examples. Each of the training example may include: (a) a distribution information of a developer on a substrate or at least one feature (e.g., location of the developer with respect to the substrate, time, relationship between the location of the developer and the time, etc.) derived from the distribution information; and (b) the classification of the distribution information or the features (e.g., whether the distribution information or the features indicate that the developer DP in the distribution information is uniform distribution or not uniform distribution). When the distribution information captured by the capturing device **251** of the analysis subsystem **250** is received as the input, the function provides an output of whether the developer DP in the distribution information is uniform distribution or not uniform distribution (i.e., the assessment result). The processor **253** of the analysis subsystem **250** can generate the assessment result regarding the uniform/non-uniform distribution of the developer DP based on the output of the function.

[0055] In some embodiments, the function is generated on an external computing hardware, and the processor **253** of the analysis subsystem **250** receives the function from said external computing hardware. In other embodiments, the distribution information captured by the capturing device **251** of the analysis subsystem **250** is transmitted to the external computing hardware. The external computing hardware applies the function to the distribution information captured by the capturing device **251** of the analysis subsystem **250** and transmits the output of the function to the processor **253** of the analysis subsystem **250**.

[0056] In some embodiments, the semiconductor device manufacturing system **200** includes a main server **300** connected to the adjustment controller **260**. The main server **300** is configured to start the developing process. In some embodiments, the main server **300** determines a sequence and timing of the developing process (i.e., the step **150**). For example, the main server **300** determines when to supply the developer DP, when to rotate the substrate **210**, when to stop rotating the

substrate **210**, and when to stop supplying the developer DP. In some embodiments, with the collaborative work of the analysis subsystem **250**, the adjustment controller **260**, and the main server **300**, the uniformity of developer coating can be dynamically controlled in real time and the abnormal coating can be detected continuously.

[0057] Referring to FIG. **3**, at step **151**, a developer on a substrate is supplied. Referring to FIGS. **2** and **3**, in some embodiments of the step **151**, the developer DP is supplied from the nozzle **240** on the substrate **210** (or on the resist layer **220**). In some embodiments, the main server **300** determines when to supply the developer DP. In some embodiments, the adjustment controller **260** applies the thrust on the pump **280** during supplying the developer DP.

[0058] At step **152**, the substrate is rotated. Referring to FIGS. **2** and **3**, in some embodiments of the step **152**, the substrate **210** is rotated by the motor **270**. For example, in FIGS. **2** and **3**, the adjustment controller **260** controls the motor **270**, the motor **270** drives the substrate support stage **230** to rotate, and the substrate **210** and the resist layer **220** are rotated accordingly to spread the developer DP. In some embodiments, the main server **300** determines when to rotate the substrate **210**. In some embodiments, the substrate **210** is rotated after supplying the developer DP on the substrate **210**.

[0059] At step **153**, a distribution information of the developer on the substrate is captured by a capturing device of an analysis subsystem. Referring to FIGS. **2** and **3**, in some embodiments of the step **153**, the distribution information of the developer DP on the substrate **210** is captured by the capturing device **251** of the analysis subsystem **250**. In some embodiments, the substrate **210** is rotated prior to capturing the distribution information of the developer DP on the substrate **210**.

[0060] At step **154**, an assessment result is generated, by a processor of the analysis subsystem, based on the distribution information. Referring to FIGS. **2** and **3**, in some embodiments of the step **154**, the assessment result is generated, by the processor **253** of the analysis subsystem **250**, based on the distribution information. In some embodiments, the assessment result is generated based on the distribution information by comparing the distribution information with the reference distribution information saved in the storage unit of the analysis subsystem **250**. In some embodiments, the distribution information of the step **153** and the assessment result of the step **154** are saved in the storage unit of the analysis subsystem **250** for training the analysis subsystem **250**.

[0061] At step **155**, whether to perform an adjustment operation based on the assessment result is determined. Referring to FIGS. **2** and **3**, in some embodiments of the step **155**, whether to perform an adjustment operation based on the assessment result is determined by the processor **253** of the analysis subsystem **250**. In some embodiments, the step **150** further includes transmitting the first control signal or the second control signal based on the assessment result from the processor **253** of the analysis subsystem **250** to the adjustment controller **260**. In some embodiments, the first control signal indicates to perform the adjustment operation, while the second control signal indicates not to perform the adjustment operation. In greater details, if it is determined to perform the adjustment operation based on the assessment result, the processor **253** of the analysis subsystem **250** demands (e.g., by transmitting the first control signal) the adjustment controller **260** to perform the adjustment operation. The adjustment operation may be performed by changing the rotational speed of the motor **270** and/or changing the thrust of the pump **280**. For example, a first rotational speed of the motor **270** in step **152** is different from a second rotational speed of the motor **270** after step **155** (e.g., performing the adjustment operation). The dispense rate of the developer DP is thus changed by changing the rotational speed of the motor **270** and/or changing the thrust of the pump **280**. On the other hand, if it is determined not to perform the adjustment operation based on the assessment result, the processor **253** of the analysis subsystem **250** demands (e.g., by transmitting the second control signal) the adjustment controller **260** not to perform the adjustment operation. In other words, the rotational speed of the motor **270** and the thrust of the pump **280** are maintained (i.e., not changed). The dispense rate of the developer DP is maintained (i.e., not changed) accordingly. In some embodiments, the step **150** further includes transmitting a warning

message corresponding to the first control signal from the processor **253** of the analysis subsystem **250** to the operators.

[0062] At step **156**, rotating the substrate is stopped. Referring to FIGS. **2** and **3**, in some embodiments of the step **156**, rotating the substrate **210** is stopped by the motor **270**. For example, in FIGS. **2** and **3**, the adjustment controller **260** controls the motor **270**, the rotational speed of the motor **270** is decreased to zero, and thus rotating the substrate **210** is stopped accordingly. In some embodiments, the main server **300** determines when to stop rotating the substrate **210**.

[0063] At step **157**, supplying the developer is stopped. Referring to FIGS. **2** and **3**, in some embodiments of the step **157**, supplying the developer DP from the nozzle **240** is stopped. In some embodiments, the main server **300** determines when to stop supplying the developer DP. In some embodiments, after stopping rotating the substrate **210**, supplying the developer DP is stopped. In some other embodiments, rotating the substrate **210** is stopped after stopping supplying the developer DP. In some embodiments, supplying the developer DP is stopped after determining whether to perform the adjustment operation based on the assessment result. In some embodiments, since capturing the distribution information of the developer DP, generating the assessment result, and determining whether to perform the adjustment operation are performed prior to stopping supplying the developer, the uniformity of developer coating can be dynamically controlled in real time and the abnormal coating can be avoided. As such, the developer DP can be more uniformly distributed (or coated) over the resist layer **220**. In some embodiments, capturing the distribution information of the developer DP, generating the assessment result, and determining whether to perform the adjustment operation are performed during supplying the developer DP. In other words, the developer DP is supplied from the nozzle **240** continuously when performing the steps **152-156** (particularly the steps **153-155**).

[0064] In some embodiments, as shown in FIGS. **1-4**, the developing process of the step **150** is performed to remove portions of the resist layer **220**. The developer DP is applied to the resist layer **220** and dissolves the portion of the resist layer **220**, leaving the developed resist layer **220** having opening(s) **225** defined therein.

[0065] Referring back to FIG. **1**, the method **100** then proceeds to step **160** after the step **150**, where a rinse process is performed. Referring to FIGS. **1** and **4**, in some embodiments of the step **160**, the rinse process is performed on the developed resist layer **220** using a basic aqueous rinse solution. In some embodiments, the rinse process is performed after the developing process, for example, to remove any residue and/or particles from the semiconductor device. In some embodiments, a rinse solution **310** is applied to the developed resist layer **220** to remove unwanted materials. The rinse solutions **310** may use organic solvent.

[0066] At step **170** in FIG. **1**, a spin dry process is performed. Referring to FIGS. **1** and **4**, in some embodiments of the step **170**, spin dry process is performed on the developed (or referred as patterned) resist layer **220** to further remove any liquids and/or solids remaining on the patterned resist layer **220** after the rinsing process. The rinsing solution **310** may be substantially removed by the spin dry process. During the spin dry process, the rinsing solution **310** is spun away from the substrate center towards the substrate edge and then removed.

[0067] At step **180** in FIG. **1**, additional fabrication operations are performed. Referring to FIGS. **1**, **5**, and **6**, in some embodiments of the step **180**, additional fabrication operations are performed on the semiconductor device using the patterned resist layer **220** as a mask. For example, the fabrication operation is applied within the opening **225** (see FIG. **5**) of the patterned resist layer **220**, while other portions of the substrate **210** covered by the patterned resist layer **220** are protected from being impacted by the fabrication operation. In some embodiments, a pattern of the opening **225** in the resist layer **220** is extended into the layer to be patterned or substrate **210** to create a pattern of opening(s) **215** in the substrate **210**, thereby transferring the pattern in the resist layer **220** into the substrate **210**. The pattern is extended into the substrate **210** by etching, using one or more suitable etchants. The resist layer **220** is at least partially removed during the etching

operation in some embodiments. In other embodiments, the resist layer **220** is removed after etching the substrate **210** by using a suitable photoresist stripper solvent or by a photoresist ashing operation. In some embodiments, a deposition process is performed to fill the opening **215** of the substrate **210** with a conductive material, thereby forming conductive lines **320** over the substrate **210** as shown in FIG. 6.

[0068] In summary, the analysis subsystem of the semiconductor device manufacturing system captures the distribution information of the developer and generates the assessment result based the distribution information, the developer can be distributed more uniformly (or evenly) over the substrate. As such, the better developing process can be achieved and a quality of the semiconductor device can be improved.

[0069] Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

[0070] It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A method of manufacturing a semiconductor device, comprising: supplying a developer on a substrate; capturing, by a capturing device of an analysis subsystem, a distribution information of the developer on the substrate; generating, by a processor of the analysis subsystem, an assessment result based on the distribution information; and determining whether to perform an adjustment operation based on the assessment result.
2. The method of claim 1, wherein performing the adjustment operation comprises changing a rotational speed of a motor, and wherein the motor is connected to a substrate support stage below the substrate.
3. The method of claim 1, wherein performing the adjustment operation comprises changing a thrust of a pump.
4. The method of claim 3, wherein the thrust of the pump is changed such that a dispense rate of the developer is changed.
5. The method of claim 1, wherein if it is determined not to perform the adjustment operation based on the assessment result, a rotational speed of a motor and a thrust of a pump are maintained, and wherein the motor is connected to a substrate support stage below the substrate.
6. The method of claim 1, wherein the assessment result is generated by comparing the distribution information with a reference distribution information saved in the analysis subsystem.
7. The method of claim 1, further comprising: transmitting a control signal based on the assessment result from the processor of the analysis subsystem.
8. The method of claim 1, further comprising: rotating the substrate prior to capturing the distribution information of the developer on the substrate.
9. The method of claim 1, further comprising: stopping supplying the developer after determining whether to perform the adjustment operation based on the assessment result.
10. The method of claim 1, wherein the distribution information of the developer on the substrate includes a surface image of the developer on the substrate.
11. The method of claim 1, further comprising: forming a resist layer over the substrate prior to supplying the developer on the substrate.
12. A semiconductor device manufacturing system, comprising: a chamber; a substrate support stage positioned in the chamber and configured to support a substrate; a nozzle positioned over the substrate support stage and configured to supply a developer on the substrate during a developing

process; and an analysis subsystem comprising: a capturing device configured to capture a distribution information of the developer on the substrate; and a processor configured to generate an assessment result based on the distribution information, wherein the processor is further configured to determine whether to perform an adjustment operation based on the assessment result.

13. The semiconductor device manufacturing system of claim 12, further comprising: an adjustment controller connected to the analysis subsystem, wherein the processor of the analysis subsystem is further configured to generate a first control signal and transmit the first control signal to the adjustment controller if it is determined to perform the adjustment operation.

14. The semiconductor device manufacturing system of claim 13, wherein the adjustment controller is configured to change a rotational speed of a motor in response to the first control signal, and wherein the motor is connected to the substrate support stage and configured to rotate the substrate.

15. The semiconductor device manufacturing system of claim 13, wherein the adjustment controller is configured to change a thrust of a pump in response to the first control signal, and wherein the pump is connected to the nozzle and configured to adjust a dispense rate of the developer from the nozzle.

16. The semiconductor device manufacturing system of claim 12, further comprising: an adjustment controller connected to the analysis subsystem, wherein the processor of the analysis subsystem is further configured to generate a second control signal and transmit the second control signal to the adjustment controller if it is determined not to perform the adjustment operation.

17. The semiconductor device manufacturing system of claim 16, wherein the adjustment controller is configured to maintain a rotational speed of a motor and a thrust of a pump in response to the second control signal, wherein the motor is connected to the substrate support stage and configured to rotate the substrate, and wherein the pump is connected to the nozzle and configured to adjust a dispense rate of the developer from the nozzle.

18. The semiconductor device manufacturing system of claim 12, wherein the processor of the analysis subsystem is configured to generate the assessment result by comparing the distribution information with a reference distribution information.

19. The semiconductor device manufacturing system of claim 12, further comprising: an adjustment controller connected to the analysis subsystem, and a main server connected to the adjustment controller, wherein the main server is configured to start the developing process.

20. The semiconductor device manufacturing system of claim 12, wherein the capturing device of the analysis subsystem includes at least one camera, and wherein the distribution information of the developer on the substrate includes a surface image of the developer on the substrate.
